MATH 547 Problem List

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Abstract

This problem list accompanies the MATH 547 course notes. The problems are split according to the corresponding sections in the notes.

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Throughout, $Z = (X, Y)$ has distribution D on $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, the sample $S = (Z_1, \dots, Z_m)$ is drawn i.i.d. from D^m, and $\mathcal{H}$ is a hypothesis class. For a loss function ℓ, write

$$L_D(h) = \mathbb{E}[\ell(h, Z)], \qquad L_S(h) = \frac{1}{m} \sum_{i=1}^{m} \ell(h, Z_i).$$

Unless another loss is specified, classification problems use the 0–1 loss.

1 Statistical learning as risk minimization

Problem 1. Represent a papaya by $x = (\text{color}, \text{softness}) \in \mathbb{R}^2$ and let $y \in \{0, 1\}$ record whether it tastes good. Let $\mathcal{H}$ be the class of axis-aligned rectangle classifiers.

1. Specify $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$.
2. Explain the distinct roles of the unknown distribution D, the observed sample S, and the class $\mathcal{H}$.
3. Write $L_D(h)$ and $L_S(h)$ under 0–1 loss.
4. Explain why $\mathcal{H}$ should be fixed before the sample is observed.

Problem 2. Let $h \in \mathcal{H}$ be fixed independently of the sample and assume $\mathbb{E}|\ell(h, Z)| < \infty$. Prove that

$$\mathbb{E}_{S \sim D^m} L_S(h) = L_D(h).$$

Then explain why this calculation alone does not imply $\mathbb{E} L_S(\widehat{h}_S) = \mathbb{E} L_D(\widehat{h}_S)$ when $\widehat{h}_S$ is selected from the data.

Problem 3. Let $\mathcal{X} = \mathcal{Y} = \{0, 1\}$ and $S = ((0, 0), (1, 1))$. Let h_0 and h_1 be the constant predictors and let $h_{\mathrm{id}}(x) = x$. Determine the empirical risk minimizers over

$$\mathcal{H}_1 = \{h_0, h_1\} \qquad \text{and} \qquad \mathcal{H}_2 = \{h_0, h_1, h_{\mathrm{id}}\}.$$

Use the result to explain why "minimize the training error" is not a complete learning rule until the admissible class is specified.

Problem 4. Let $A(S)$ be the output of a learner and put $L_D^* = \inf_{h \in \mathcal{H}} L_D(h)$.

1. State the PAC target under the assumption $L_D^* = 0$.
2. State the agnostic PAC target without a realizability assumption.
3. If $L_D^* = 0.18$ and $\varepsilon = 0.03$, what risk threshold must an agnostic learner meet with probability at least $1 - \delta$?
4. Explain why a guarantee below L_D^* cannot hold uniformly for a learner constrained to output a member of $\mathcal{H}$.

Problem 5. Let $\mathcal{X}$ be a discrete domain. For $z \in \mathcal{X}$, define $h_z(x) = \mathbf{1}\{x = z\}$ and let $h_-(x) = 0$. Consider

$$\mathcal{H}_{\text{sing}} = \{h_z : z \in \mathcal{X}\} \cup \{h_-\}.$$

Assume realizability.

1. Give an ERM algorithm for this class.
2. Prove that if $m \geq \log(1/\delta)/\varepsilon$, its output has risk at most ε with probability at least $1 - \delta$.
3. Explain why this example does not follow from the finite-class theorem when $\mathcal{X}$ is infinite.

Problem 6. Suppose an algorithm A satisfies, for every distribution D,

$$L_D(A(S)) \leq \inf_{h \in \mathcal{H}} L_D(h) + \varepsilon$$

with probability at least $1 - \delta$ whenever $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$. Prove that the same algorithm and sample-complexity bound satisfy the ordinary PAC requirement on every distribution realizable by $\mathcal{H}$.

2 Finite hypothesis classes

Problem 7. Let $\mathcal{H}$ be finite, assume realizability, and let $\widehat{h}_S$ be an ERM. For $\varepsilon > 0$, define

$$\mathcal{H}_{\text{B}} = \{h \in \mathcal{H} : L_D(h) > \varepsilon\}, \qquad M_h = \{S : L_S(h) = 0\}.$$

Prove

$$\{S : L_D(\widehat{h}_S) > \varepsilon\} \subseteq \bigcup_{h \in \mathcal{H}_{\text{B}}} M_h.$$

Identify the exact step at which realizability is used.

Problem 8. Fix $h \in \mathcal{H}$ with $L_D(h) = p$ and assume the examples are independent.

1. Prove that $\mathbb{P}(L_S(h) = 0) = (1 - p)^m \leq e^{-mp}$.
2. Combine this fact with Problem 7 and the union bound to show

$$\mathbb{P}(L_D(\widehat{h}_S) > \varepsilon) \leq |\mathcal{H}| e^{-m\varepsilon}.$$

3. Explain why the events M_h need not be independent.

Problem 9. Let $|\mathcal{H}| = 50$, $\varepsilon = 0.05$, and $\delta = 0.01$.

1. Compute the smallest integer m guaranteed by the realizable bound

$$m \geq \frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}.$$

2. Compute the smallest integer m guaranteed by the agnostic bound

$$m \geq \frac{2\log(2|\mathcal{H}|/\delta)}{\varepsilon^2}.$$

3. Explain the different dependence on ε.

Problem 10. Call S α-representative if

$$\sup_{h \in \mathcal{H}} |L_S(h) - L_D(h)| \leq \alpha.$$

Assume a true-risk minimizer $h^* \in \arg\min_{h \in \mathcal{H}} L_D(h)$ exists. Prove that if S is $(\varepsilon/2)$-representative, every ERM $\widehat{h}_S$ satisfies

$$L_D(\widehat{h}_S) \leq L_D(h^*) + \varepsilon.$$

Label the three inequalities in the proof according to the fact that justifies each one.

Problem 11. Assume $\ell(h, Z) \in [0, 1]$ for every $h \in \mathcal{H}$ and $\mathcal{H}$ is finite. Using Hoeffding's inequality for a fixed h and then a union bound, prove

$$\mathbb{P}\left(\sup_{h \in \mathcal{H}} |L_S(h) - L_D(h)| > \alpha \right) \leq 2|\mathcal{H}| e^{-2m\alpha^2}.$$

Set $\alpha = \varepsilon/2$ and combine the result with Problem 10 to recover the finite-class agnostic ERM bound.

Problem 12. Let $\mathcal{H}$ be finite and suppose $\ell(h, Z) \in [a, b]$ almost surely for every $h \in \mathcal{H}$. Derive a sufficient sample size for agnostic ERM of the form

$$m \geq \frac{2(b-a)^2 \log(2|\mathcal{H}|/\delta)}{\varepsilon^2}.$$

Your proof should state the version of Hoeffding's inequality being used and should explain why the ERM transfer step is unchanged.

Problem 13. In the proof of the finite-class agnostic guarantee, identify precisely where each of the following is used:

1. the i.i.d. sampling assumption;
2. boundedness of the loss;
3. finiteness of $\mathcal{H}$; and
4. existence of a true-risk minimizer.

For the final item, explain how the proof can be rewritten with an η-minimizer when the infimum is not attained.

3 No-Free-Lunch and inductive bias

Problem 14. Fix a learner A, choose $C \subseteq \mathcal{X}$ with $|C| = 2m$, and draw the unlabeled sample $(X_1, \dots, X_m)$ uniformly from C^m. Let U be the set of points in C absent from the sample. A target $F : C \to \{0, 1\}$ is then chosen uniformly from all labelings, and $Y_i = F(X_i)$.

1. Prove that $|U| \geq m$.
2. Conditional on the labeled sample, prove that for each $u \in U$, the probability over F that $A(S)(u) \neq F(u)$ is $1/2$.
3. Deduce that the expected risk, averaged over targets and samples, is at least $1/4$.
4. Explain why one fixed target f must have expected risk at least $1/4$.

Problem 15. Let R be a random variable taking values in $[0, 1]$ and suppose $\mathbb{E}R \geq 1/4$. Prove

$$\mathbb{P}(R \geq 1/8) \geq 1/7.$$

Apply the result to $R = L_D(A(S))$ to complete the final step of the No-Free-Lunch proof.

Problem 16. Write the main logical content of No-Free-Lunch in the form "for every A there exists D such that ...." Then write the statement obtained by reversing the first two quantifiers. Explain why the theorem proves the first statement but not the second, and why the hard distribution may depend on the learner.

Problem 17. Let $\mathcal{X}$ be infinite and let $\mathcal{H} = \{h : \mathcal{X} \to \{0, 1\}\}$. Assume for contradiction that $\mathcal{H}$ is PAC learnable. Choose $\varepsilon < 1/8$ and $\delta < 1/7$, then use the No-Free-Lunch theorem on a $2m$-point subset of $\mathcal{X}$ to derive a contradiction for $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$.

Problem 18. Suppose $\mathcal{H}_2$ consists of axis-aligned rectangle classifiers built from two features, while $\mathcal{H}_5$ uses five features, with $\mathcal{H}_2 \subseteq \mathcal{H}_5$.

1. Compare their best achievable risks and explain which class can have smaller approximation error.
2. Compare the difficulty of controlling their estimation errors at a fixed sample size.
3. Explain how your preference should change as the number of labeled examples increases.

Do not assert that one class is always preferable; state the tradeoff and the information needed to decide.

Problem 19. Let $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \cdots$ and suppose that, on an event $\mathcal{E}$, for every k and every $h \in \mathcal{H}_k$,

$$|L_S(h) - L_D(h)| \leq b_k.$$

Choose $(\widehat{k}, \widehat{h})$ to minimize $L_S(h) + b_k$ over all pairs with $h \in \mathcal{H}_k$. Prove that on $\mathcal{E}$,

$$L_D(\widehat{h}) \leq \inf_k \left\{ \inf_{h \in \mathcal{H}_k} L_D(h) + 2b_k \right\}.$$

Interpret the two terms on the right as approximation and estimation.

4 VC dimension

Problem 20. For $a < b$, define

$$h_a(x) = \mathbf{1}\{x < a\}, \qquad h_{a,b}(x) = \mathbf{1}\{a < x < b\}.$$

Prove that the threshold class has VC dimension 1 and the interval class has VC dimension 2. Each proof must include both a shattered set and an argument that no larger set is shattered.

Problem 21. Let $\mathcal{H}$ be the class of axis-aligned rectangles in $\mathbb{R}^2$.

1. Explain why displaying one particular five-point set that is not shattered does not prove $\mathrm{VCdim}(\mathcal{H}) \le 4$.
2. For an arbitrary five-point set, select points attaining the left, right, bottom, and top extremes. Construct a labeling that no rectangle realizes and conclude the upper bound.
3. Give a four-point configuration that is shattered and conclude $\mathrm{VCdim}(\mathcal{H}) = 4$.

Problem 22. Let $\mathcal{H}' \subseteq \mathcal{H}$ be binary hypothesis classes on the same domain. Prove

$$\mathrm{VCdim}(\mathcal{H}') \le \mathrm{VCdim}(\mathcal{H}).$$

Give an example in which the inequality is strict and an example in which $\mathcal{H}' \subsetneq \mathcal{H}$ but the VC dimensions are equal.

Problem 23. Let $\mathcal{X}$ be a finite set of size n and, for $0 \le k \le n$, let

$$\mathcal{H}_k = \{\mathbf{1}_A : A \subseteq \mathcal{X},\ |A| = k\}.$$

Prove

$$\mathrm{VCdim}(\mathcal{H}_k) = \min\{k, n-k\}.$$

For the lower bound, show how an arbitrary labeling of a suitable set can be extended to a k-element subset of $\mathcal{X}$. For the upper bound, use both the all-zero and all-one labelings.

Problem 24. For $I \subseteq \{1, \dots, n\}$ define

$$h_I(x) = \left(\sum_{i \in I} x_i\right) \bmod 2, \qquad x \in \{0,1\}^n.$$

Prove that the class $\{h_I : I \subseteq \{1, \dots, n\}\}$ has VC dimension n. Use the standard basis for the lower bound and the finite-class VC bound for the upper bound.

Problem 25. Let

$$\mathcal{H}_d = \{x \mapsto \mathbf{1}\{\langle w, x\rangle > 0\} : w \in \mathbb{R}^d\}.$$

Prove $\mathrm{VCdim}(\mathcal{H}_d) = d$. For the lower bound, use the standard basis. For the upper bound, begin with a linear dependence $\sum_{i=1}^{d+1} \alpha_i x_i = 0$ and choose a labeling from the signs of the coefficients. Treat zero coefficients explicitly.

Problem 26. Let $\mathcal{H}_d^{\text{rec}}$ be the class of indicators of axis-aligned rectangles in $\mathbb{R}^d$.

1. Show that the $2d$ points $\{\pm e_1, \ldots, \pm e_d\}$ are shattered. For each coordinate, choose an interval containing neither, one, or both of -1 and 1 while always containing 0.
2. Given any $2d+1$ points, select a point attaining the minimum and maximum in each coordinate. Use an unselected point to construct an impossible labeling.
3. Conclude that $\text{VCdim}(\mathcal{H}_d^{\text{rec}}) = 2d$.

Problem 27. Assume $\text{VCdim}(\mathcal{H}) = \infty$. For an arbitrary proposed sample size m, choose a set of $2m$ points shattered by $\mathcal{H}$. Explain why every target labeling in the No-Free-Lunch construction is realized by a member of $\mathcal{H}$, and use this to prove that $\mathcal{H}$ is not PAC learnable.

5 Growth functions

Problem 28. Let $C = \{c_1 < \cdots < c_m\} \subset \mathbb{R}$.

1. Count the distinct restrictions to C of threshold classifiers and prove that their growth function is $m + 1$.
2. Count the nonempty consecutive blocks of C, include the empty labeling, and prove that the interval growth function is $1 + m(m+1)/2$.
3. Check that each formula equals 2^m up to the corresponding VC dimension and is strictly smaller afterward.

Problem 29. For the class $\mathcal{H}_{\text{sing}} = \{h_z : z \in \mathcal{X}\} \cup \{h_-\}$ from Problem 5, assume $\mathcal{X}$ is infinite. Determine $\text{VCdim}(\mathcal{H}_{\text{sing}})$ and compute $\tau_{\mathcal{H}_{\text{sing}}}(m)$ exactly for every $m \geq 1$.

Problem 30. Evaluate

$$\sum_{i=0}^{2} \binom{5}{i}.$$

Compare it with the exact number of interval labelings on five ordered points. Is Sauer's bound tight in this case?

Problem 31. Let $C = \{c_1, \ldots, c_m\}$ and $C' = C \setminus \{c_1\}$. Let Y_0 be the restrictions to C' that occur with at least one label at c_1, and let Y_1 be those that occur with both labels at c_1.

1. Prove $|\mathcal{H}_C| = |Y_0| + |Y_1|$.
2. If $\text{VCdim}(\mathcal{H}) \leq d$, prove that the class represented by Y_1 has VC dimension at most $d - 1$.
3. Derive
$$T(m, d) \leq T(m-1, d) + T(m-1, d-1)$$
and close the induction using Pascal's identity.

Problem 32. Let $\mathcal{X}$ be infinite and let

$$\mathcal{C}_k = \{A \subseteq \mathcal{X} : |A| = k\}.$$

Regarding $\mathcal{C}_k$ as a class of indicator functions, prove that every set of at most k points is shattered and no set of $k+1$ points is shattered. Then show that on every m-point set the number of induced labelings is

$$\sum_{j=0}^{\min\{k,m\}} \binom{m}{j}.$$

Conclude that the Sauer bound is attained.

Problem 33. Let $F : S \to T$ be a function and let $\mathcal{C}$ be a class of subsets of T. Define

$$F^{-1}(\mathcal{C}) = \{F^{-1}(C) : C \in \mathcal{C}\}.$$

Prove

$$\mathrm{VCdim}(F^{-1}(\mathcal{C})) \leq \mathrm{VCdim}(\mathcal{C}).$$

If a finite set $A \subseteq S$ is shattered by the pullback class, first show that F is injective on A and then prove that $F(A)$ is shattered by $\mathcal{C}$.

Problem 34. Let $\mathcal{H} = \bigcup_{j=1}^{r} \mathcal{H}_j$ and suppose $\mathrm{VCdim}(\mathcal{H}_j) \leq d$ for every j, where $1 \leq d < m$. Prove

$$\tau_{\mathcal{H}}(m) \leq r \sum_{i=0}^{d} \binom{m}{i} \leq r \left(\frac{em}{d}\right)^d.$$

Deduce that if $\mathcal{H}$ shatters an m-point set, then necessarily

$$2^m \leq r \left(\frac{em}{d}\right)^d.$$

Explain how this inequality limits the VC dimension of a finite union.

Review Problems (Sections 1–5)

Problem 35. Let $S = \{(x_i, y_i) : i = 1, \ldots, m\} \subseteq \mathbb{R} \times \{0, 1\}$ have distinct instances and labels generated by a function. Define the memorizing rule

$$h_S(x) = \begin{cases} y_i, & x = x_i \text{ for some } i, \\ 0, & \text{otherwise.} \end{cases}$$

Construct a real polynomial p_S such that $h_S(x) = \mathbf{1}\{p_S(x) \geq 0\}$ for every $x \in \mathbb{R}$. Conclude that ERM over all thresholded polynomials can overfit.

Problem 36. Let $\mathcal{H}^{\mathrm{circ}} = \{h_r : r \geq 0\}$ on $\mathbb{R}^2$, where $h_r(x) = \mathbf{1}\{\|x\|_2 \leq r\}$. Under realizability, describe a consistent ERM and prove that

$$m_{\mathcal{H}^{\mathrm{circ}}}(\varepsilon, \delta) \leq \frac{\log(1/\delta)}{\varepsilon}.$$

Your argument should identify the annular error region and compute the probability that an i.i.d. sample misses it.

Problem 37. Let $\mathcal{H}$ be a finite binary class and let $f \in \mathcal{H}$ be the realizable target. Suppose $X_i \sim D_i$ independently, although the distributions need not be identical, and put $\overline{D}_m = m^{-1} \sum_{i=1}^{m} D_i$. Prove that, for every $\varepsilon \in (0, 1)$,

$$\Pr\Big\{\exists h \in \mathcal{H} : L_{(\overline{D}_m, f)}(h) > \varepsilon,\ L_S(h) = 0\Big\} \leq |\mathcal{H}| e^{-m\varepsilon}.$$

Use the arithmetic–geometric mean inequality and state where independence is needed.

Problem 38. For a distribution D on $\mathcal{X} \times \{0, 1\}$, define $\eta(x) = \Pr(Y = 1 \mid X = x)$ and $f_D(x) = \mathbf{1}\{\eta(x) \geq 1/2\}$. Prove that

$$L_D(f_D) \leq L_D(g)$$

for every classifier $g : \mathcal{X} \to \{0, 1\}$. Make the comparison pointwise conditional on $X = x$, and then integrate.

Problem 39. Let A be any learning algorithm, let D be any distribution, and assume the loss takes values in $[0, 1]$. Prove that the following are equivalent:

1. for every $\varepsilon, \delta > 0$ there is $m(\varepsilon, \delta)$ such that, for every $m \geq m(\varepsilon, \delta)$, $\Pr\{L_D(A(S)) > \varepsilon\} < \delta$;
2. $\lim_{m \to \infty} \mathbb{E}[L_D(A(S))] = 0$.

Prove one direction using boundedness and the other using Markov's inequality.

Problem 40. Let A be a binary-classification learner and suppose $|\mathcal{X}| \geq km$ for an integer $k \geq 2$. Strengthen the No-Free-Lunch construction to show that there is a realizable distribution D for which

$$\mathbb{E}_{S \sim D^m} L_D(A(S)) \geq \frac{1}{2} - \frac{1}{2k}.$$

Identify the unseen portion of a km-point support and average over all target labelings before fixing one target.

Problem 41. Let $\mathcal{H}' \subseteq \mathcal{H}$ be binary hypothesis classes on the same domain. Prove $\mathrm{VCdim}(\mathcal{H}') \leq \mathrm{VCdim}(\mathcal{H})$. Give one strict example and one proper inclusion for which equality holds.

Problem 42. Let $\mathcal{X}$ be finite and $0 \leq k \leq |\mathcal{X}|$. Determine, with proof, the VC dimensions of

$$\mathcal{H}_{=k} = \{h \in \{0, 1\}^{\mathcal{X}} : |\{x : h(x) = 1\}| = k\}$$

and

$$\mathcal{H}_{\mathrm{edge,k}} = \{h : |\{x : h(x) = 1\}| \leq k \text{ or } |\{x : h(x) = 0\}| \leq k\}.$$

Include both the shattering constructions and the upper bounds.

Problem 43. In the proof of Sauer's lemma one obtains, for a finite set A and a class $\mathcal{H}$ of VC dimension d,

$$|\mathcal{H}_A| \le |\{B \subseteq A : B \text{ is shattered by } \mathcal{H}\}| \le \sum_{i=0}^{d} \binom{|A|}{i}.$$

Give examples realizing all four possible patterns obtained by making each of the two displayed inequalities either an equality or a strict inequality.

Problem 44. Let $\mathcal{H}_1, \dots, \mathcal{H}_r$ be binary classes and put $d = \max_j \mathrm{VCdim}(\mathcal{H}_j) \ge 3$. Prove

$$\mathrm{VCdim}\left(\bigcup_{j=1}^{r} \mathcal{H}_j\right) \le 4d \log(2d) + 2 \log r.$$

Begin with a set of q points shattered by the union, use Sauer's lemma to bound its number of labelings by rq^d, and solve the resulting inequality.

6 Uniform convergence and the Fundamental Theorem

Problem 45. Let $S = (Z_i)_{i=1}^m$ and $S' = (Z_i')_{i=1}^m$ be independent samples from D, and let $\ell(h, z) = \mathbf{1}\{h(x) \ne y\}$.

1. Prove $\mathbb{E}_S \sup_h |L_D(h) - L_S(h)| \le \mathbb{E}_{S,S'} \sup_h |L_{S'}(h) - L_S(h)|$.
2. Introduce independent Rademacher signs by swapping the pairs (Z_i, Z_i'). Justify equality of the expected suprema before and after the swaps.
3. Conditional on S, S', show that only $\tau_{\mathcal{H}}(2m)$ loss vectors need to be considered. For each fixed vector prove the tail bound $2e^{-m\rho^2/2}$ for the signed average. Explain where the interval length enters this exponent.

Problem 46. Suppose $\mathrm{VCdim}(\mathcal{H}) = d \ge 1$. You may use $\tau_{\mathcal{H}}(2m) \le (2em/d)^d$ for $2m > d + 1$ and the uniform bound

$$\sup_h |L_D(h) - L_S(h)| \le \frac{8 + 2\sqrt{\log \tau_{\mathcal{H}}(2m)}}{\delta \sqrt{2m}}$$

with probability at least $1 - \delta$. Prove that $\mathcal{H}$ has uniform convergence. Then derive the corresponding excess-risk bound for an ERM, and identify the dependence on m, d, δ that this particular proof gives.

Problem 47. For a finite binary hypothesis class $\mathcal{H}$, one has $\mathrm{VCdim}(\mathcal{H}) \le \lfloor \log_2 |\mathcal{H}| \rfloor$.

1. Find an infinite class of functions from $[0, 1]$ to $\{0, 1\}$ with VC dimension 1.
2. Give a finite binary hypothesis class on $[0, 1]$ for which $\mathrm{VCdim}(\mathcal{H}) = \lfloor \log_2 |\mathcal{H}| \rfloor$. Prove the equality.
3. *Extension.* Use the Fundamental Theorem to determine whether each class you constructed has uniform convergence and is agnostically PAC learnable under 0–1 loss. Explain why infinite cardinality alone does not prevent learnability.

Problem 48. On $\mathcal{X} = \mathbb{R}$, consider the one-parameter binary class

$$\mathcal{H} = \{h_\theta : \theta \in \mathbb{R}\}, \qquad h_\theta(x) = \mathbf{1}\{\sin(\theta x) > 0\}.$$

Prove that $\mathrm{VCdim}(\mathcal{H}) = \infty$.

Hint. Use the points $2, 2^2, \ldots, 2^d$. For a number $t = 0.b_1 b_2 \cdots$ in binary with a nonterminating, non-eventually-constant tail, the sign of $\sin(2^j \pi t)$ is determined by b_j.

Extension. Deduce that $\mathcal{H}$ is not PAC learnable and does not have uniform convergence under 0–1 loss. Explain why the number of real parameters need not bound the VC dimension.

Problem 49. Let $\mathcal{H} \subseteq \{0,1\}^{\mathcal{X}}$ satisfy $\mathrm{VCdim}(\mathcal{H}) \geq d \geq 1$. Fix a sample size $m \geq 1$ and any learning rule A that returns a binary classifier from m labeled examples. Prove that there exists a distribution D on $\mathcal{X} \times \{0,1\}$, realizable by $\mathcal{H}$, such that

$$\mathbb{E}_{S \sim D^m} L_D(A(S)) \geq \min_{h \in \mathcal{H}} L_D(h) + \frac{d-m}{2d}.$$

The distribution may depend on A and m. If A is randomized, include its internal randomness in the expectation.

Hint. Put uniform mass on a shattered set of d points, average over its 2^d possible target labelings, and consider points absent from the sample before selecting a fixed target labeling.

Problem 50. Let $\mathcal{H}_1, \mathcal{H}_2 \subseteq \{0,1\}^{\mathcal{X}}$ be binary hypothesis classes on the same domain, and set $d = \max\{\mathrm{VCdim}(\mathcal{H}_1), \mathrm{VCdim}(\mathcal{H}_2)\}$. Assume $3 \leq d < \infty$. Prove the sharper two-class union bound

$$\mathrm{VCdim}(\mathcal{H}_1 \cup \mathcal{H}_2) \leq 2d + 1.$$

Hint. If a set of $2d+2$ points were shattered by the union, apply Sauer's lemma to each class and compare $2\sum_{j=0}^{d} \binom{2d+2}{j}$ with 2^{2d+2}.

Extension. Deduce that ERM over $\mathcal{H}_1 \cup \mathcal{H}_2$ is an agnostic PAC learner. Use the quantitative Fundamental Theorem to give an upper bound, up to an absolute constant, on the sample size needed for excess risk at most ε with probability at least $1 - \delta$. The benchmark should be the smaller of the two infimal population risks, $\min_{i=1,2} \inf_{h \in \mathcal{H}_i} L_D(h)$.

Problem 51. Let $\mathcal{A}$ be a class of subsets of $\mathcal{X}$, let P be a probability measure, and let S be an ε-net: every $A \in \mathcal{A}$ with $P(A) \geq \varepsilon$ intersects S. For a fixed realizable binary target h^*, take $\mathcal{A} = \{\{x : h(x) \neq h^*(x)\} : h \in \mathcal{H}\}$. Prove that every hypothesis agreeing with h^* on S has population error less than ε. Prove also that the VC dimension of this disagreement class equals that of $\mathcal{H}$.

7 Rademacher complexity and generalization

For a nonempty bounded set $A \subseteq \mathbb{R}^m$, write

$$R(A) = \frac{1}{n}\mathbb{E}_\varepsilon \sup_{a\in A} \sum_{i=1}^m \varepsilon_i a_i,$$

where the ε_i are independent rademacher random variables.

Problem 52. For fixed vectors $a_1, \dots, a_N \in \mathbb{R}^m$ and independent signs ε_i, prove

$$R := \frac{1}{m}\mathbb{E}_\varepsilon \max_{j\le N} \sum_{i=1}^m \varepsilon_i a_{j,i} \le \frac{\max_j \|a_j\|_2 \sqrt{2\log N}}{m},$$

In other words, prove Massart's lemma.

Problem 53. Let $\mathcal{F}$ be a nonempty class of Boolean functions $f : \Omega \to \{0,1\}$, and let $X, X_1, \dots, X_m$ be independent random points in Ω with the same distribution. Prove

$$\mathbb{E}\sup_{f\in\mathcal{F}} \left(\frac{1}{m}\sum_{i=1}^m f(X_i) - \mathbb{E}f(X) \right) \le 2\mathbb{E}\sup_{f\in\mathcal{F}} \frac{1}{m}\sum_{i=1}^m \varepsilon_i f(X_i),$$

Problem 54. Let X' be an independent copy of X.

1. If $X \sim \mathrm{Ber}(p)$, with $\mathbb{P}(X=1) = p = 1 - \mathbb{P}(X=0)$, compute the probability mass function of $X - X'$.
2. If X has density $e^{-x}\mathbf{1}\{x \ge 0\}$, prove that $X - X'$ has density $\frac{1}{2}e^{-|x|}$ on $\mathbb{R}$.

Problem 55. Let $X_1, \dots, X_N$ be independent integrable random vectors in $\mathbb{R}^d$, equipped with any norm $\|\cdot\|$. Let $\varepsilon_1, \dots, \varepsilon_N$ be independent signs, independent also of these vectors. The vectors need not have mean zero.

1. Prove
$$\mathbb{E}\left\| \sum_{i=1}^N (X_i - \mathbb{E}X_i) \right\| \le 2\mathbb{E}\left\| \sum_{i=1}^N \varepsilon_i X_i \right\|.$$
2. Explain why there is no universal constant $c > 0$ for which the reverse inequality
$$c\mathbb{E}\left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \le \mathbb{E}\left\| \sum_{i=1}^N (X_i - \mathbb{E}X_i) \right\|$$
holds for all such vectors.

Problem 56. Let $T \subseteq \mathbb{R}^m$ be nonempty and bounded, and let $\varphi_i : \mathbb{R} \to \mathbb{R}$ satisfy $|\varphi_i(u) - \varphi_i(v)| \le |u - v|$. Prove

$$\mathbb{E}_\varepsilon \sup_{t\in T} \sum_{i=1}^m \varepsilon_i \varphi_i(t_i) \le \mathbb{E}_\varepsilon \sup_{t\in T} \sum_{i=1}^m \varepsilon_i t_i$$

by following these steps.

1. For $T \subseteq \mathbb{R}^2$ and a 1-Lipschitz function φ, verify
$$\begin{aligned} &\sup_{t\in T}(t_1 + \varphi(t_2)) + \sup_{t\in T}(t_1 - \varphi(t_2)) \\ &\qquad \le \sup_{t\in T}(t_1 + t_2) + \sup_{t\in T}(t_1 - t_2). \end{aligned}$$
2. Use induction on m to complete the proof.

Problem 57. Generalize the preceding contraction inequality to functions $\varphi_i : \mathbb{R} \to \mathbb{R}$ with arbitrary finite Lipschitz constants $L_i \geq 0$. State and prove an inequality that makes the dependence on the L_i explicit, without assuming $\varphi_i(0) = 0$. Here $T \subseteq \mathbb{R}^m$ remains nonempty and bounded, and the signs are independent.

8 Concentration Inequalities: An Introduction

Problem 58. Let X and Y be real random variables with the same mean. Suppose $X \in [a, b]$ almost surely and $Y \in \{a, b\}$ almost surely, where $a < b$. Prove

$$\mathbb{E}e^{\lambda X} \leq \mathbb{E}e^{\lambda Y} \qquad \text{for every } \lambda \in \mathbb{R}.$$

Problem 59. Let $X \in [a, b]$ almost surely. Prove Hoeffding's lemma,

$$\mathbb{E}e^{\lambda(X-\mathbb{E}X)} \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right) \qquad (\lambda \in \mathbb{R}),$$

using the following cumulant argument.

1. For $a < b$, reduce to the case $\mathbb{E}X = 0$, $b - a = 1$, and $X \in \{a, b\}$ almost surely. Treat $a = b$ separately.
2. In the reduced case, compute $K(\lambda) = \log \mathbb{E}e^{\lambda X}$ and verify $K(0) = K'(0) = 0$ and $K''(\lambda) \leq 1/4$. Deduce $K(\lambda) \leq \lambda^2/8$ for every real λ.

Problem 60. Let $X_1, \ldots, X_N$ be independent, with $a_i \leq X_i \leq b_i$ almost surely. Assume $V = \sum_{i=1}^N (b_i - a_i)^2 > 0$. Deduce from Hoeffding's lemma that

$$\mathbb{P}\left\{\sum_{i=1}^N (X_i - \mathbb{E}X_i) \geq t\right\} \leq \exp\left(-\frac{2t^2}{V}\right) \qquad (t > 0).$$

Problem 61. Let $\mathcal{H}$ be a nonempty finite hypothesis class and let $\ell : \mathcal{H} \times \mathcal{Z} \to [a, b]$, where $a < b$. The sample is i.i.d., and an ERM rule minimizes L_S over $\mathcal{H}$. For $\epsilon > 0$ and $\delta \in (0, 1)$, let $m_{\mathcal{H}}^{\mathrm{UC}}(\eta, \delta)$ denote a distribution-free sample size ensuring $\sup_{h \in \mathcal{H}} |L_D(h) - L_S(h)| \leq \eta$ with probability at least $1 - \delta$, and let $m_{\mathcal{H}}(\epsilon, \delta)$ denote the corresponding agnostic sample complexity for ERM. Prove

$$m_{\mathcal{H}}(\epsilon, \delta) \leq m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon/2, \delta) \leq \left\lceil \frac{2(b-a)^2 \log(2|\mathcal{H}|/\delta)}{\epsilon^2} \right\rceil.$$

Problem 62. For a fixed input to a decision problem, a randomized algorithm returns the correct answer with probability $1/2 + \epsilon$, where $0 < \epsilon \leq 1/2$. Run it N times with independent randomness and return the majority answer; any tie may be resolved arbitrarily. Prove that, for every $\delta \in (0, 1)$, the answer is correct with probability at least $1 - \delta$ whenever

$$N \geq \frac{1}{2\epsilon^2} \log \frac{1}{\delta}.$$

9 Subgaussian and subexponential random variables

For the concentration problems, use $\|X\|_{\psi_2} = \inf\{a > 0 : \mathbb{E}e^{X^2/a^2} \le 2\}$ and $\|X\|_{\psi_1} = \inf\{a > 0 : \mathbb{E}e^{|X|/a} \le 2\}$. The letters $c, C > 0$ denote absolute constants, which may change between inequalities.

Problem 63. If $|X| \le B$ almost surely, prove $\|X\|_{\psi_2} \le B/\sqrt{\log 2}$. For $g \sim N(0, \sigma^2)$, compute $\mathbb{E}e^{g^2/a^2}$ and deduce $\|g\|_{\psi_2} = \sqrt{8/3}\,\sigma$. Explain why a subgaussian variable need not be centered, and why the bound $\mathbb{E}e^{\lambda X} \le e^{C\lambda^2\|X\|_{\psi_2}^2}$ for all real λ does require centering.

Problem 64. Prove that a real random variable X is subgaussian if and only if there is a constant $K > 0$ such that

$$\mathbb{E}\exp(\lambda^2 X^2) \le \exp(\lambda^2 K^2) \qquad \text{whenever } |\lambda| \le 1/K.$$

More precisely, show that if X is subgaussian and $\|X\|_{\psi_2} > 0$, this inequality holds with $K = \|X\|_{\psi_2}$. Interpret the zero-norm case separately. Conversely, prove that the displayed inequality implies $\|X\|_{\psi_2} \le 2K$. Note: this is NOT the same thing as in the equivalent characterization of subgaussian random variables (4) since in that setting X is centered, the exponent in the LHS is λX and the inequality holds for all λ.

Problem 65. Let X and Y be independent, mean-zero, subgaussian random variables. Prove that there are absolute constants $c, C > 0$ such that

$$c(\|X\|_{\psi_2} + \|Y\|_{\psi_2}) \le \|X + Y\|_{\psi_2} \le C(\|X\|_{\psi_2} + \|Y\|_{\psi_2}).$$

Problem 66. By modifying the proof of the equivalent subgaussian properties, prove the equivalence of the following three properties for a real random variable X:

1. There is $K_1 > 0$ such that $\mathbb{P}\{|X| \ge t\} \le 2e^{-t/K_1}$ for all $t \ge 0$.
2. There is $K_2 > 0$ such that $\|X\|_{L^p} := (\mathbb{E}|X|^p)^{1/p} \le K_2 p$ for every $p \ge 1$.
3. There is $K_3 > 0$ such that $\mathbb{E}e^{|X|/K_3} \le 2$.

Show that the parameters in these implications differ by at most absolute constant factors.

Problem 67. State and prove a subexponential version of the centering inequality for subgaussian norms. In particular, prove that every subexponential random variable X satisfies

$$\|X - \mathbb{E}X\|_{\psi_1} \le C\|X\|_{\psi_1}$$

for an absolute constant $C > 0$.

Problem 68. Let $X_1, \dots, X_N$ be independent, mean-zero, subexponential random variables, and set

$$\sigma^2 = \sum_{i=1}^N \|X_i\|_{\psi_1}^2, \qquad K = \max_{1 \le i \le N} \|X_i\|_{\psi_1}.$$

Use the subexponential Bernstein inequality to show that, for an absolute constant $C > 0$ and every $u \ge 0$,

$$\mathbb{P}\left\{\sum_{i=1}^N X_i \ge C(\sigma\sqrt{u} + Ku)\right\} \le 2e^{-u},$$

provided $\sigma > 0$. When $\sigma = 0$, state the corresponding conclusion using a strict upper-tail event.

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